DEPI | DATA ANALYSIS TRACK

Requirements Gathering

Stakeholders, user stories, use cases, functional and non-functional requirements, and analytical questions

Air Quality and Life Expectancy: Exploring the Relationship Between Air Pollution and Population Health

Team Members

Ganat Saad Abdullah Hussein Ahmed | **Team Leader**
Shaimaa Maghraby Mohamed Houssien Elagouz
Habiba Maged Abdel Baqi
Dina Farag Mohamed Mohamed
Nada Ramzy Mohamed Ghnam

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1. Project Context

The project will combine the supplied WHO Ambient Air Quality Database (2018-version workbook) with the Life Expectancy dataset to create a country-year analytical view. The intended output is an exploratory report/dashboard supported by documented data preparation and analysis using Excel/Power Query, Power BI, and SQL if used by the team.

2. Stakeholder Analysis

Stakeholder	Information need	How the project may support it
Public-health researchers and students	Understand patterns between particulate pollution and population-health indicators.	Show descriptive associations and comparisons with coverage caveats.
Environmental and air-quality analysts	Compare available PM2.5/PM10 observations across countries and years.	Summarize city observations at country-year level and show contributing city counts.
Public-health policy teams	Identify patterns that may merit further investigation.	Provide transparent evidence summaries and clearly state data limitations.
Data analysts and decision-makers	Understand data quality, coverage, and how results were produced.	Document cleaning, definitions, validation counts, and calculations.
DEPI instructor and project evaluators	Assess whether the team can frame the problem, prepare data, analyze it, and communicate results.	Provide the project documentation, reproducible analysis, dashboard, and presentation.

3. User Stories

- As a public-health researcher, I want to compare PM2.5 and life expectancy across matched country-year records so that I can identify patterns for further investigation.
- As an environmental analyst, I want to view average PM10 and PM2.5 alongside the number of cities contributing to each average so that I can judge the available coverage.
- As a policy analyst, I want to compare results by year and development status so that I can explore whether patterns differ between groups when enough data are available.
- As a data-quality reviewer, I want to see the matched record count and missing-value coverage so that I can assess the reliability and limits of the results.
- As a project evaluator, I want the report values to match the cleaned analytical table so that I can verify the calculations.

4. Use Cases

Use case	Primary actor	Typical steps	Expected result
Explore pollution and life expectancy	Researcher / student	Open the report; select a year or range; review the scatter plot and summary values.	Displays the observed relationship for matched records.
Compare development groups	Analyst	Select Developed/Developing where available; compare summary statistics and charts.	Shows group-level patterns with sample sizes.
Assess data coverage	Data reviewer	Inspect matched country-year count, city count, missingness, and year coverage.	Makes uneven coverage and sparse observations visible.
Review country-year details	Analyst	Select a country and year using available filters; inspect pollutant and health values.	Displays available matched values and contributing city counts.
Validate reported results	Project team / instructor	Compare report totals and averages with the validated output table.	Confirms that report calculations agree with the analysis table.

5. Functional Requirements

- FR1. Import the two source workbooks while preserving the original files.
- FR2. Standardize field names, data types, and country identifiers; document manual country-name mappings.
- FR3. Create cleaned numeric PM10 and PM2.5 fields while preserving raw values and conversion notes/flags.
- FR4. Aggregate valid city observations to country-year averages and calculate the number of distinct cities contributing to each pollutant average.
- FR5. Merge air-quality summaries with life-expectancy records using validated country identifiers and year.
- FR6. Identify duplicate keys, unmatched records, missing values, year coverage, and final matched sample size.
- FR7. Calculate descriptive summaries for PM10, PM2.5, life expectancy, and adult mortality; calculate correlations only on valid paired records.
- FR8. Create suitable visuals for pollution distributions, pollutant-health scatter plots, year comparisons, and development-status comparisons when sample sizes are adequate.
- FR9. Provide relevant filters such as Year, Country, and Status where those fields are available.
- FR10. Display source definitions and limitations, including city coverage and the fact that association does not prove causation.

6. Non-Functional Requirements

- NFR1. Accuracy: displayed totals, averages, and counts must reconcile with the validated analytical table.
- NFR2. Traceability: raw values, cleaned values, transformation rules, and country-name mappings must be documented.
- NFR3. Usability: report pages must have clear titles, units, legends, and readable labels.
- NFR4. Performance: filters and visuals should remain responsive with the final merged dataset.
- NFR5. Transparency: comparisons should show or allow users to inspect the available sample size and coverage.
- NFR6. Responsible interpretation: the report must not describe correlation as proof of causation or city averages as complete national exposure estimates.
- NFR7. Reproducibility: transformations and calculation definitions should be retained in the workbook/query steps or project notes.

7. Business / Analytical Questions

1. Across matched country-year observations, is average PM2.5 associated with life expectancy?
2. Is average PM10 associated with adult mortality in the available matched records?
3. How do average PM2.5 and PM10 vary across the years represented in the final merged dataset?
4. Do observed pollution-life expectancy patterns differ between Developed and Developing countries where sufficient observations exist?
5. How does life expectancy vary with GDP in the matched sample?
6. How does life expectancy vary with schooling in the matched sample?
7. Which country-year records have the highest and lowest available PM2.5 and PM10 values, and what are their city-coverage counts?
8. How many country-year observations are successfully matched, and how does the match count vary by year?
9. What proportion of key pollution and health fields is missing after merging?
10. Do the main descriptive patterns change when converted PM values are excluded, if enough measured-only records remain?

Source datasets:

- Life Expectancy Data: https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who
- Ambient Air Quality Database: https://www.kaggle.com/datasets/fkosmowski/ambient-air-quality-database
- WHO Air Quality Database, 2018 update: https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database/2018